DS51M07 Global Shutter Camera Module Specification

Product Name: USB Camera Module

Model: DS51M07B

Version: V1.1

1. Overview

This camera module series uses a B&W CMOS 1MP (1280×800) image sensor based on OmniPixel®3-GS technology and a high-performance controller. With advanced circuit design, it features high image quality, low distortion, and low noise.

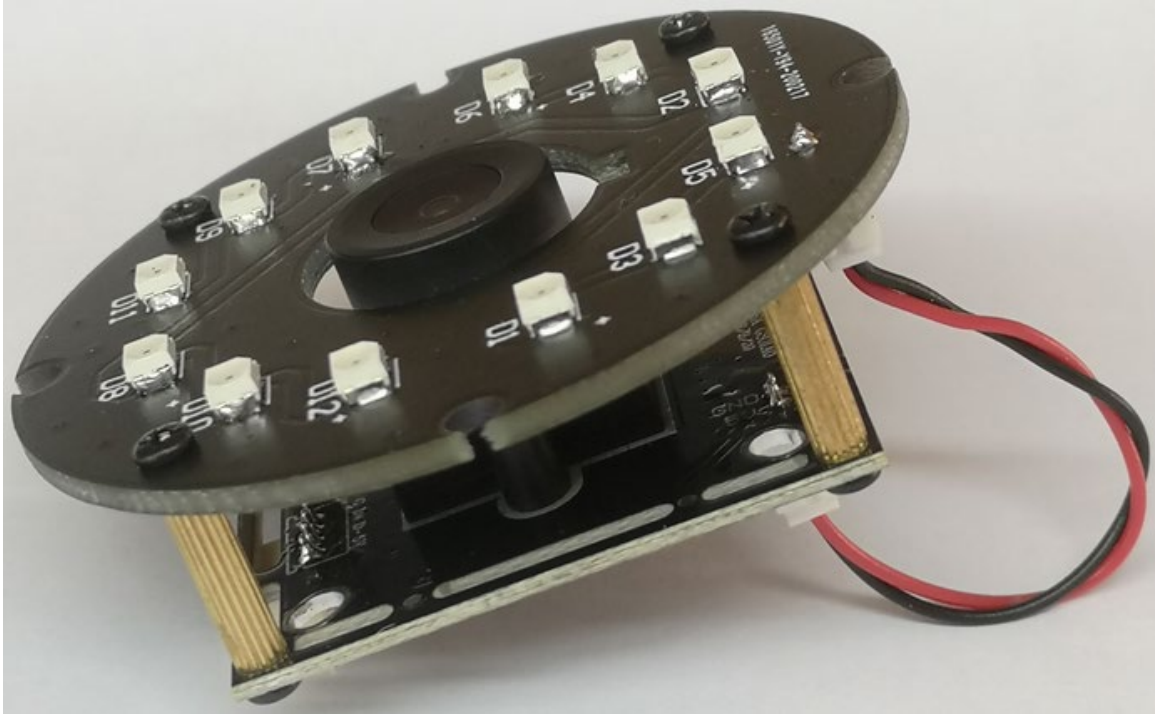
Key Features:

- 1MP 1/4-inch low power image sensor
- Resolution up to 720p
- Excellent low-light performance
- Global shutter with automatic electronic shutter
- Automatic gain control
- USB2.0 plug and play
- YUV / MJPEG output supported

2. Product Appearance

Standard size: 38mm × 38mm

Surface shall be free of scratches, cracks, deformation and contamination.



3. Camera Parameters

Image Sensor: OV9281

Sensor Size: 1/4” CMOS

Shutter: Global Shutter

Frame Rate: 120fps

Resolution: 1280×800

Min Illumination: ≥0.1Lux @F1.2

Dynamic Range: ≥80dB

Interface: USB2.0

Power: 5V±5%

Power Consumption: Max 0.6W

Lens: M12, f=3.1mm

FOV: D90° H74° V45°

Operating Temp: -20°C~+60°C

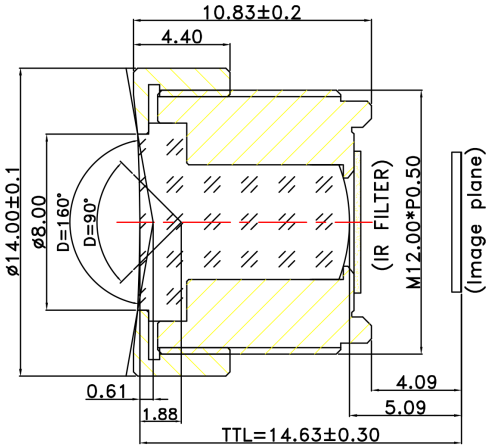
Humidity: 15~85%RH

PRODUCT NAME : 3137-A1

1. SPECIFICATION:

1.RESOLUTION 5M
2.MAX IMAGE CIRCLE ø6.4 mm
3.FOCAL LENGTH (EFL) f= 3.1±5% mm
4.F/NO (INFINITE) F/NO= 2.8±5%
5.BACK FOCAL LENGTH BFL= 5.09±0.2 mm
6.FLANGE BACK LENGTH FB=4.09±0.2 mm
7.Thread Size M12*P0.5
8.Element 5G
9.IR FILTER SPEC.(Bulit-in,Others available)
Tavg>=90% @ 440-620 nm
T=50% @ 650±10 nm
Tavg<=3% @ 700-1000 nm
T<5% @ 1050 nm

NOTE		SPECIFICATION	
Sensor		1/3" OV4689	1/4" OV9281
FOV	Vertical	60°	45°
	Horizontal	116°	74°
	Diagonal	160°	90°
DISTORTION		<-28%	<-27%
Relative Ill.		>96%	>97%
CRA		<17°	<12.7°



2. OPTICAL LAYOUT:

REV	DESCRIPTION	DATE	NAME	NAME	3137-A1			SCALE		DRAWN	
①	新规作成	2018.8.10		VIEW		REV	00				
②								DATE		CHECKED	
③											
④								BORDER		APPROVED	

4. Environmental

Operating Temp: -20~+60℃

Storage Temp: -40~+90℃

Humidity: 15%~85%

MTBF: ≥50000h

5. Electrical

Rated Voltage: 5V

Voltage Range: 4.2~5.5V

Max Current: 190mA

6. Communication

Interface: USB2.0

Speed: 480Mbps

Cable: 4 core shielded cable

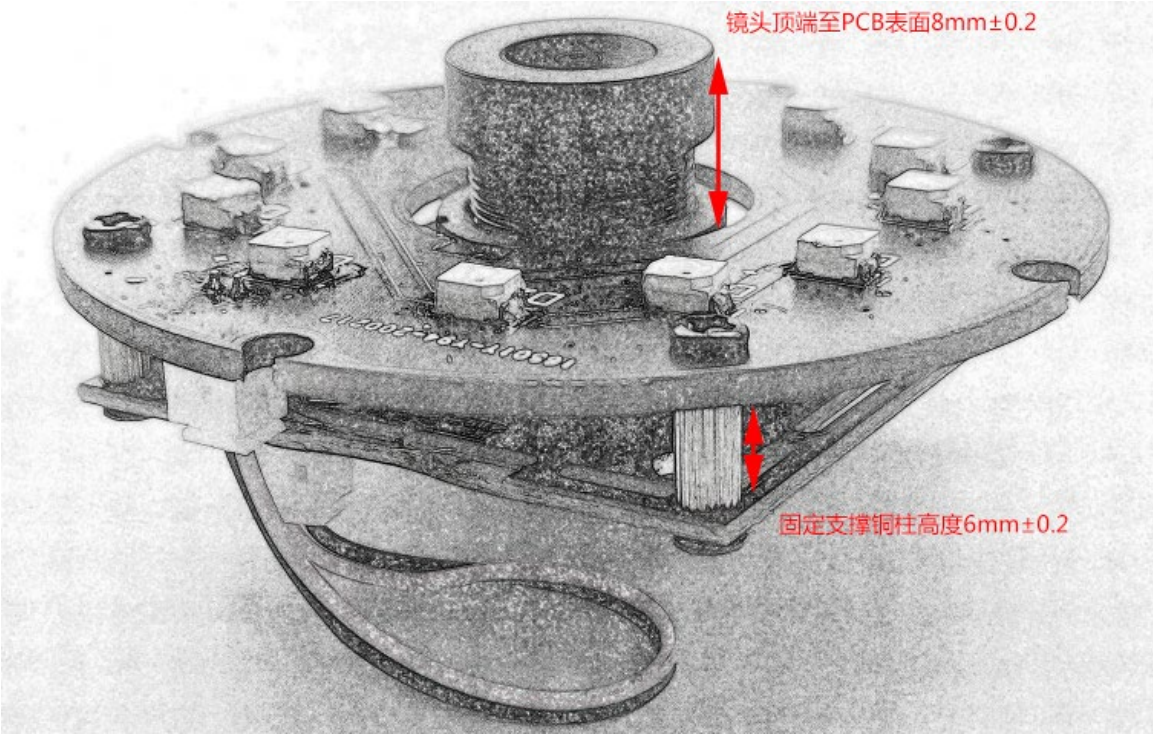
Protocol: UVC

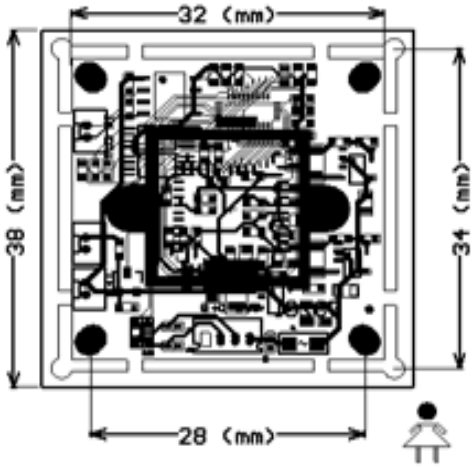
7. Mechanical

Module Size: 38mm × 38mm

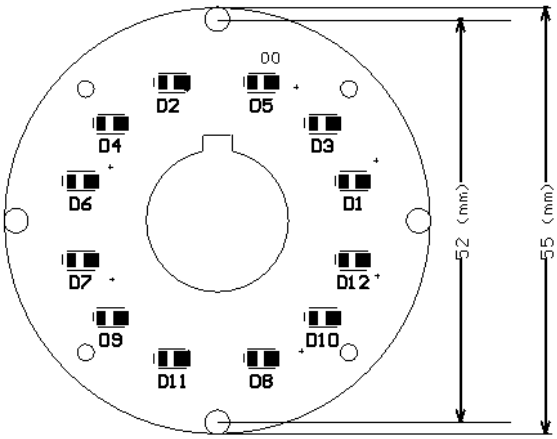
Max Size: 55×55×22.5mm

Mounting: Light board fixing

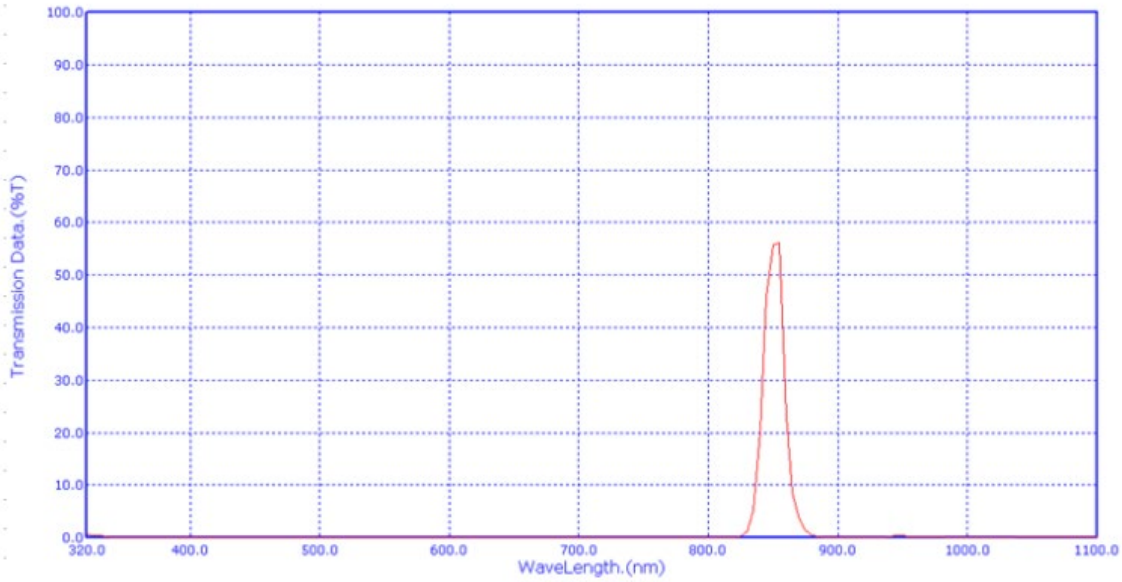

镜头顶端至PCB表面8mm±0.2
固定支撑铜柱高度6mm±0.2


32 (mm)
38 (mm)
34 (mm)
28 (mm)

15.5mm
1.2mm


D0
D2
D5
D4
D3
D6
D1
D7
D12
D9
D10
D11
D8
52 (mm)
55 (mm)

8. Filter



850nm narrow band filter

Center wavelength: 850nm

Bandwidth: 840~860nm